Rebuilding the Coverage Gap Score from Public Reference Data, and Unnamed Local Roads on Tribal Reservations: Methodology for the Bias Bounty Mapping Equity Challenge

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October 30, 2026

Abstract

The challenge asks for a coverage gap score per Census tract that compares Overture Maps release 2026-08-19.0 with reference data for roads, buildings and points of interest across 9,794 tracts in five hazard regions of the western and southern United States. No labels are provided, so every submission is an estimate of a deterministic formula applied to reference layers that the organisers removed from the challenge bucket on 25 September 2026. Over sixteen submissions an estimation approach built only from the challenge package brought the public RMSE from 0.1072 for the all-zero file to 0.0745. After Zindi staff confirmed on 27 October that the public versions of those layers could be used, the pipeline rebuilt the reference side of each component from those sources and reached 0.02996 (submission S19), 0.02380 (S24) once ZIP establishment counts were apportioned to tracts by HUD business-address ratios, and 0.02152 (S25) once the building reference was rebuilt from Microsoft's Global ML footprints. The rebuilt transport component reproduces the organisers' published undefined rates to within one percentage point in all four documented regions, and their published spread of road length ratios (0.71 to 1.59) once road length on shared tract boundaries is counted once. The submissions that tested alternative building, EMS, school and establishment references show where the remaining error sits. For the Best Bias Discovery track, this paper reports that on federally recognised reservations the share of local street length with no name in Overture is 9.5 percentage points higher than on non-tribal tracts of the same region and rurality (95 % county bootstrap interval 3.5 to 14.7), rising to 27.6 points on the Osage Reservation, while highways there are named at the same rate as elsewhere. The scorecard cannot see this because it measures length and counts. Emergency dispatch and evacuation messaging depend on the names.

1 Introduction

The Bias Bounty Mapping Equity Challenge, run on Zindi by Humane Intelligence with Reliabl and Radiant Earth, asks how complete open map data is in places exposed to heat, drought and wildfire, and whether the gaps fall unevenly across rural, tribal and socially vulnerable communities. Each scored tract receives a composite score $y \in [0, 1]$, with 0 meaning Overture matches the reference

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and 1 meaning Overture has nothing where the reference has something. Ranking is by root mean squared error (RMSE) on y, with about 30 % of tracts on the public leaderboard and 70 % on the private one.

The score is a formula. It is the mean of up to three components (transport, building, points of interest), each comparing an Overture quantity with a reference quantity in the same tract. The Overture side is in the challenge package. The reference side was removed from the package on 25 September 2026 (TIGER/Line roads, Microsoft building footprints, HIFLD and USGS facilities, and Census Business Patterns establishments), so the task becomes the estimation of a formula whose inputs are partly hidden.

This paper documents the pipeline that produced the final submissions, how each component was computed, the edge cases, the alternatives that were tried and why each was kept or dropped, and a bias pattern that the automated Bias Scorecard cannot detect. Section 2 lists the data. Section 3 describes the pipeline. Section 4 defines each component and how it was computed. Section 5 covers the edge cases. Section 6 gives the full submission history with the evidence each one produced. Section 7 reflects on the scorecard and Section 8 presents the bias discovery.

All submissions were made by Zindi user `craigouma`, entering solo. The two submissions selected for private scoring are S25 (public RMSE 0.021520, Zindi ID `2ncgdcQQ`) and S24 (0.023796, `EC3xBxjr`), described in Section 6.3; the others referred to in this paper are listed in Table 4.

2 Data

2.1 Challenge package

All challenge data were read from the challenge's public Source Cooperative bucket [1] on 25 October 2026. The bucket held 327 objects; 172 files (4,615,029,100 bytes) were synced to Kaggle, every byte count was checked against the listing, and all matched on the first attempt. No file named for a coverage gap, TIGER, Microsoft, HIFLD or Census Business Patterns existed anywhere in the bucket.

The pipeline uses the six Overture layers of release 2026-08-19.0 per region (buildings, roads, unfiltered roads, rail, infrastructure, places), the ACS 2020 to 2024 housing unit counts per tract, the tract polygons and the strata tables (Census tract table, CDC SVI 2022, Climate Vulnerability Index, RUCA 2020, RUCC 2023, tribal overlap, and the drought, heat and wildfire tables). GEOID is kept as 11-character text in every read; an integer cast drops the leading zero of Arizona's state code and silently breaks every Maricopa join. Region membership comes from the `region` column of the sample submission, which places tract `35023970000` (Hidalgo County, New Mexico) in the Maricopa region.

Two uses of the package should be disclosed. ACS housing units, which the organisers describe as a sanity check outside the building ratio, were used as a feature in the estimation phase (S02 to S16). Overture's own `sources[].dataset` attribute, which records whether a footprint came from OpenStreetMap or a machine-learned product, was used in the same phase. Neither enters the final submissions.

2.2 Public reference sources

On 27 October 2026 Zindi staff confirmed that the public versions of the removed layers (TIGER/Line roads, Microsoft footprints, HIFLD facilities and Census Business Patterns) could be used. Facility counts were taken from the USGS National Map structures, which the challenge page names as the facility reference and which is the source of the HIFLD fire station layer.

The pipeline then downloaded the sources in Table 1 directly from their publishers inside Kaggle kernels. The one exception is the HUD USPS ZIP-TRACT crosswalk [9], which HUD now serves only through a token API. Its archived copy on DataLumos (project 219325, version V2) was downloaded on 29 October 2026, the Q4 2024 ZIP-TRACT file was converted from xlsx to parquet with ZIP and tract codes kept as text (`scripts/hud_dataset.py`), and the result was published as a private Kaggle dataset read by stage 07. No earlier version of the challenge bucket, no old challenge zip and no third-party mirror of the removed layers was used. Organiser score files of any kind were never opened; a file guard rejects any path containing `coverage-gap` or `coverage_gap` regardless of other permissions, and a unit test fails if any reader bypasses it.

Table 1: Reference sources rebuilt from their publishers. "Final" marks the sources used by the selected submissions S25 and S24.

Source	Vintage	Used for	Final
Census TIGER/Line primary and secondary roads (S1100, S1200)	2025	transport reference	S25, S24
USGS National Map structures, fire stations (FCODE 74026)	2026	fire reference	S25, S24
USGS National Map structures, schools (FCODE 73002 to 73005)	2026	school reference	S25, S24
Census ZIP Code Business Patterns, establishments per ZIP	2023	establishments reference	S25, S24
HUD USPS ZIP-TRACT crosswalk, business-address ratios	Q4 2024	apportioning ZIP counts	S25, S24
Census 2020 ZCTA to tract relationship file	2020	apportioning ZIP counts (S20)	no
USGS National Map structures, ambulance services (FCODE 74034)	2026	EMS reference (S21)	no
Microsoft Global ML Building Footprints	2026-02-03	building reference	S25
Microsoft US Building Footprints	v2	building reference (S18)	no
HIFLD Open archive: fire and EMS stations, public and private schools	2025	S22	no

TIGER/Line is distributed in NAD83 longitude and latitude and the tract polygons, built from the same TIGER edges, are labelled CRS84. The pipeline relabels TIGER as CRS84 without a transform, which keeps road and tract boundary vertices identical; NAD83 and WGS84 differ by under two metres in the contiguous United States.

3 Pipeline

Every step that reads challenge data or fits anything runs as a private Kaggle script kernel; the laptop is used only for editing, unit tests on synthetic fixtures and orchestration through the Kaggle command line. Code reaches Kaggle as a private dataset built from a clean git commit, and each kernel stops if the commit hash in the dataset differs from the one it was rendered with. Figure 1 shows the stages and Table 2 their measured cost.

Spatial work uses DuckDB 1.5.4 with its spatial extension. Two of its behaviours fail silently and are guarded in code and tests: transforming from EPSG:4326 without `always_xy` returns infinite coordinates, and `ST_Length_Spheroid` on longitude and latitude order returns NaN, so every length passes through a helper that flips coordinates, asserts a finite result and is tested against `pyproj.Geod` to within 0.1 %. Lengths are geodesic metres of each line clipped to the tract polygon.

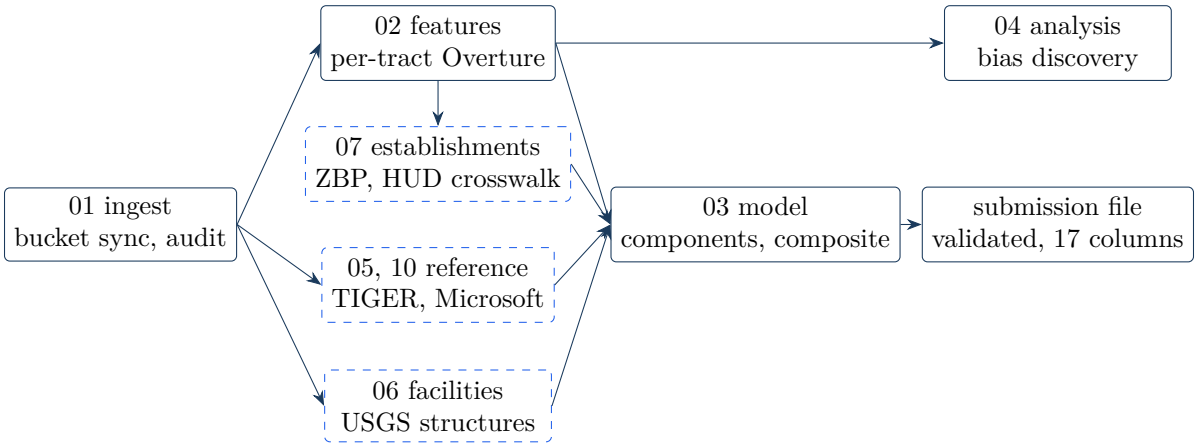


Figure 1: Kaggle stages. Solid boxes use the challenge package only; dashed boxes download public reference sources. Stage 08 (HIFLD archive) and stage 09 (TIGER boundary lengths) are diagnostics outside the final path.

Table 2: Stage cost on Kaggle CPU sessions (4 cores, 32.9 GB RAM), from the runs that fed the final submissions and the bias analysis. Wall time includes queueing.

Stage	Run	Wall (min)	Peak RSS (GB)
01 ingest	20261025-1434-01_ingest-398b1ba	5.4	1.39
02 features	20261026-1136-02_features-c1f6404	27.9	6.99
05 reference	20261027-2145-05_reference-0add93b	33.9	3.55
06 facilities	20261028-0024-06_facilities-8d8b042	8.7	0.49
07 establishments	20261030-0946-07_establishments-decc7f9	1.2	0.22
10 Global ML	20261030-1127-10_globalml-2221625	33.8	2.13
03 model (S25)	20261030-1227-03_model-45a6fcc	1.7	0.56
03 model (S24)	20261030-0947-03_model-30d804e	1.4	0.55
04 analysis	20261030-1019-04_analysis-8a838bb	24.3	0.80
04 analysis (S25 map)	20261030-1514-04_analysis-7cec170	21.6	0.80

4 Components

4.1 Score definition

For an Overture quantity O and a reference quantity R in the same tract,

$$\mathrm{gap}(O, R) = \max\left(0,\ 1 - \frac{O}{R}\right), \qquad \text{defined only when } R > 0. \tag{1}$$

The composite is the mean of the defined components among transport T, building B and points of interest P,

$$y = \frac{\sum_{c \in \{T,B,P\}} \delta_c \, c}{\sum_c \delta_c}, \tag{2}$$

where δ_c is 1 when component c is defined. P is the mean of its defined halves, a facilities half F and an establishments half E, and F is the mean of the defined per-type gaps for fire stations, EMS stations and schools. Tracts with every component undefined were removed from the scored list by the organisers, so every scored tract has at least one defined component. An undefined component is written as 0 with its flag false.

4.2 Transport

O is the length of Overture road segments of class motorway, trunk, primary or secondary, clipped to the tract. R is the length of TIGER/Line 2025 roads with MTFCC S1100 or S1200, clipped the same way. A TIGER road that runs along a tract boundary intersects both neighbouring tracts and is counted in full in each, as a clipped computation would count it. T is undefined where the tract holds no S1100 or S1200 length.

Two published facts check this rebuild without the leaderboard (Table 3). The share of tracts with an undefined component was published per region; building and POI are almost never undefined, so it should match the share with no TIGER length, and it does to within one percentage point in all four regions. The organisers also published that the ratio of Overture to TIGER length ranges from 0.71 to 1.59 across the four regions. Summing clipped per-tract lengths gives 0.64 to 1.29. Counting the TIGER length within 0.1 m of a tract boundary a single time gives 0.73 for Eastern Oklahoma and 1.58 for South-Central Texas, the two ends of the published range. The organisers' region totals therefore count each road once while their per-tract values count boundary roads in both tracts, which is the rule used here.

Table 3: Transport checks against published values. Undefined: share of scored tracts with no TIGER S1100 or S1200 length. Ratio: Overture named-class length over TIGER length, summed over the region, with boundary TIGER length counted once.

Region	Undefined	Published	Ratio	Mean T where defined
Eastern Oklahoma	0.212	0.21	0.731	0.355
Northern California	0.369	0.37	1.248	0.162
South-Central Texas	0.284	0.28	1.581	0.123
Maricopa	0.546	0.55	1.314	0.165
Eastern Washington	0.248	n/a	0.965	0.199

4.3 Building

O is the count of Overture footprints whose centroid falls in the tract and R the Microsoft footprint count, also by centroid. The public Microsoft US Building Footprints v2 release, rebuilt per tract (S18), gives a mean B of 0.003 and raised the public RMSE from 0.0491 to 0.0546. The centroid counting was checked independently: the Overture and US v2 counts per tract equal a competitor's published counts from the same public files in all 415 Eastern Washington tracts. The organisers' release is therefore a different one. Microsoft's Global ML Building Footprints [10], release 2026-02-03, downloaded by stage 10 as the zoom-9 quadkey files covering each region, holds 6 to 17 % more footprints than US v2 in every region. Used as R in S25, it makes B positive in 31 % of tracts with a mean of 0.0056, changes 5,644 tracts, and lowered the public RMSE from 0.0238 to 0.0215; the line search along S24 to S25 gave $t^* = 1.22$. B is defined wherever Global ML has a footprint in the tract. S24 instead sets $B = 0$ wherever Overture has footprints and $B = 1$ where it has none and the tract holds housing units, with the level of B the one leaderboard-tuned scalar it uses (a constant baseline gap tested at 0.01 and 0.05 in S13, with 0 chosen).

4.4 Points of interest

The organisers' bucket README states how Overture places are matched: `categories.primary` equal to `fire_department` for fire stations, `ambulance_and_ems_services` for EMS, and `elementary_school`, `middle_school`, `high_school`, `school`, `private_school` or `public_school` for schools. Each per-type gap applies Equation 1 to the Overture count and the reference count, and is undefined where the tract holds no reference facility of that type.

Facilities half. The reference counts are USGS National Map structure points in the tract: FCODE 74026 for fire stations and 73002 to 73005 for schools. The README states that EMS is close to one to one with its reference and rarely binds, and that fire stations carry the facility half. Adding EMS against USGS ambulance services (74034) made the public RMSE worse (S21), and a line search between S19 and S21 put the best mix at 7 % of the EMS change, so EMS is left out of the facility mean. Replacing USGS schools with the HIFLD public and private school layers (S22) carried no signal (best mix 0.2 %).

Establishments half. The reference is Census Business Patterns establishments. No tract-level count exists, so the organisers must have apportioned county or ZIP counts to tracts, and their method is undocumented. Places in Overture outnumber ZIP Business Patterns establishments by 2.0 to 2.2 to 1 in every region, so E is 0 in almost every tract under any apportionment. S19 sets $E = 0$ where the tract has residents. S20 apportions ZIP Business Patterns 2023 counts to tracts through the 2020 ZCTA to tract relationship file, weighting each tract's share of a ZCTA by its Overture place count as a stand-in for business addresses, which makes E positive in 1.4 % of tracts. A line search between S19 and S20 put the best mix at 24 % of the S20 change: some establishment gaps exist, placed differently from this apportionment.

The standard tool for moving ZIP-level business counts to tracts is the HUD USPS ZIP-TRACT crosswalk, which gives for each ZIP the share of its business addresses (`BUS_RATIO`) and residential addresses (`RES_RATIO`) in each tract. A competitor's public repository, posted to the challenge discussion, lists the organisers' establishment reference columns as `cbp_estab` and `cbp_estab_res`, which match these two ratios; no file from that repository was used. S24 therefore sets the reference to ZIP Business Patterns 2023 establishments times `BUS_RATIO` from the Q4 2024 crosswalk, summed over the ZIPs that reach each tract, compares it with all Overture places in the tract,

and leaves E undefined where the result is 0. E becomes positive in 2.4 % of tracts and 253 tracts change. The public RMSE fell from 0.0300 to 0.0238, and the line search along S19 to S24 gave $t^* = 1.17$, so the change is right in full. Offline checks of the 2023 crosswalk quarters, ZBP 2022, the residential and total ratios, and scale factors from 0.9 to 1.3 found no variant clearly better than S24.

4.5 Composite and definedness

With every component rebuilt from counts and lengths, definedness follows directly from the reference counts, and the composite in Equation 2 is computed directly. The estimation phase (S02 to S16) instead drew 2,000 Monte Carlo samples per tract of each component's definedness and value, dropped draws with all three undefined to respect the conditioning on scored tracts, and averaged the composite over the remaining draws. That code remains in the pipeline and reduces to the direct computation when every definedness probability is 0 or 1.

5 Edge cases

Zero-population tracts. Population enters no formula. In S25 and S24, E is defined wherever the apportioned establishment count is positive, which holds in 22 of the 27 scored tracts without residents (airports, parks, industrial land); its gap is 0 in all of them because Overture holds more places than the apportioned count. S19 leaves E undefined in these tracts. The facility gaps are unaffected because they depend on facility points, which can exist in a tract without residents.

Water-dominated tracts. Seven Texas water tracts (codes 99xx) were dropped from the scored list by the organisers. Two remain, `48039990000` (Brazoria County) and `48057990000` (Calhoun County). Neither has residents, a TIGER primary or secondary road, a USGS fire station or school, an Overture place or an Overture footprint. The Global ML file holds one footprint in the first and 48 in the second, likely offshore and shoreline structures, so the building component is defined in both; with no Overture footprints its gap is 1. Both are written with $B = 1$ and a composite of 1, which is also the only value consistent with a scored tract having a defined component.

Tracts with no Overture data in a layer. The Overture count or length is 0, so the gap is 1 wherever the reference is positive and the component is undefined where it is not. No imputation is applied to the Overture side.

Cross-border tracts. The New Mexico tract `35023970000` belongs to the Maricopa region and the Oklahoma border tracts `40013795900` and `40085094200` to Eastern Oklahoma. TIGER and USGS files were downloaded for every state that holds a scored tract (Arizona, New Mexico, California, Oklahoma, Texas, Washington), so these tracts are treated like any other.

Eastern Washington. The 415 Eastern Washington tracts appear in the sample submission but not in the organisers' published rates. They are processed with the same method; with rebuilt references no parameter needs to be borrowed from other regions.

6 Submission history

Every submission was registered before upload with a hypothesis, the change against the current best and the expected direction. Table 4 lists them and Figure 2 plots the scores.

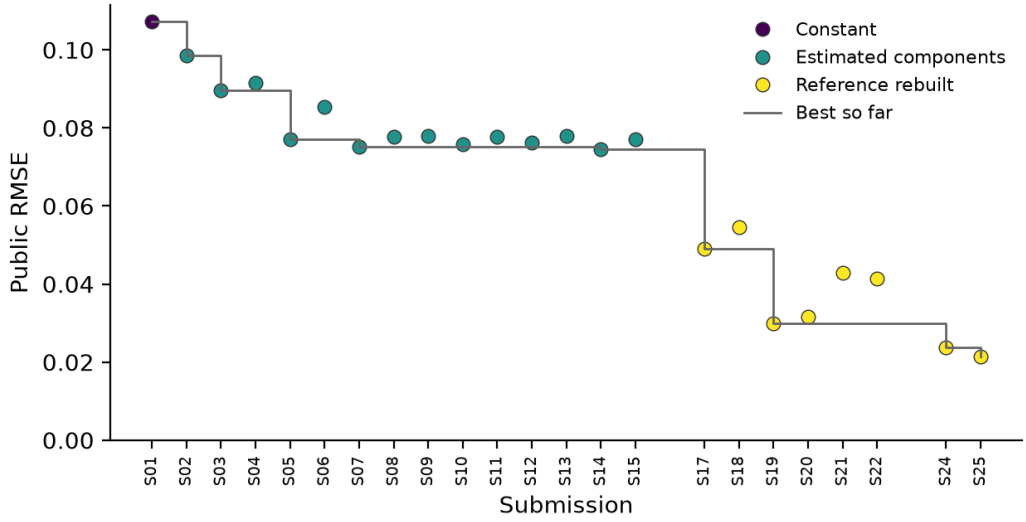


Figure 2: Public RMSE by submission. S16 was not scored; S23, a competitor's file uploaded as a benchmark, is left out.

6.1 Reading a submission without labels

Each public score fixes one inner product with the hidden target, and two submissions together say how far to move from one towards the other. For a current best p and a candidate $q = p + d$ with public RMSEs R_p and R_q,

$$E[(y-p)\,d] = \tfrac{1}{2}\left(R_p^2 - R_q^2 + E[d^2]\right), \qquad t^* = \frac{E[(y-p)\,d]}{E[d^2]}, \tag{3}$$

where t^* is the step along d that would minimise the error. A value near 1 means the candidate's change is right in full, near 0 that it carries no signal, and negative that it points the wrong way. This was used as a diagnostic on whole-method pairs, to decide whether a hypothesis was right (S19 from S17, $t^* = 0.93$), carried no signal (S22, 0.002) or pointed the wrong way (S18, -0.84). No final submission is a fitted blend, and no per-tract or per-region edit was ever driven by a score.

6.2 Tuned scalars

The project allowed at most six global scalars tuned on the public leaderboard. Five were used during the estimation phase: the boundary distance (25 m from 0, 25, 50), the POI level (0.779), the transport value scale (1.15), the building baseline gap (0 from 0.01 and 0.05) and the fire model strength (0.3). S25 uses none of them; S24 uses only the building baseline gap.

6.3 Final selection

Figure 3 maps the S25 scores.

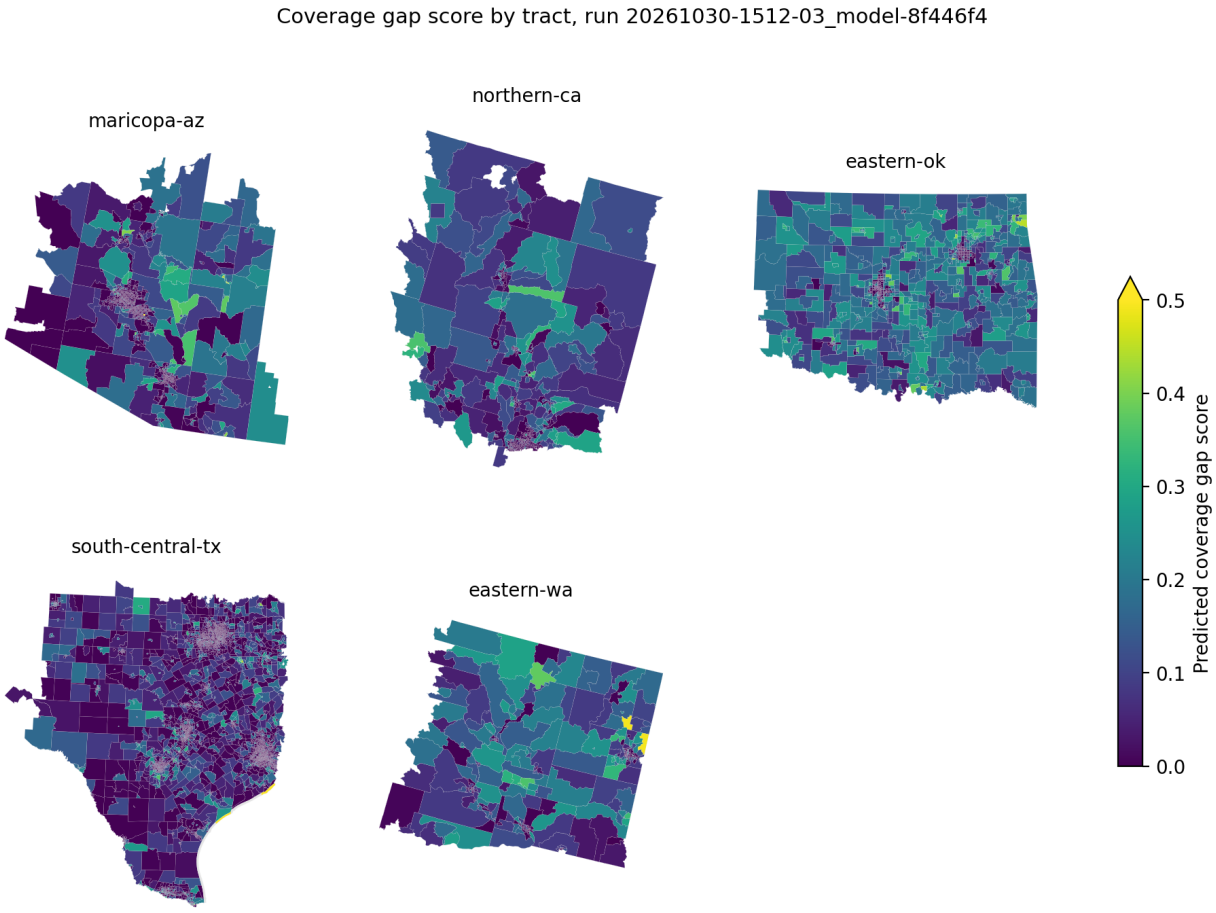


Figure 3: S25 coverage gap score by tract, drawn by stage 04 from the S25 reproduction run (analysis run `20261030-1514-04_analysis-7cec170`). Scores above 0.5 share the top colour.

Table 4: Every scored submission, what it tested and what was decided.

ID	Change tested	Public RMSE	Decision
S01	Sample submission, all zero	0.10725	Fixes $E[y^2]$ for later diagnostics
S02	Analytic v1: route-network transport proxy, building and POI estimates	0.09849	Base
S03	Boundary double counting of the transport reference, $d = 25$ m	0.08974	Kept
S04	Same with $d = 50$ m	0.09151	25 m kept
S05	POI as a regional constant	0.07710	POI variation carried little
S06	Building reference from USFS building counts	0.08545	Dropped
S07	Establishments half at 0	0.07518	Kept
S08	Lower transport reference noise	0.07782	Dropped
S09	Tract-level facility half	0.07805	Dropped
S10	Transport reference without ramps	0.07590	Dropped
S11	Unclipped segment lengths	0.07766	Dropped
S12	Transport level probe	0.07632	Scale 1.15
S13	Building level probe	0.07801	Baseline 0
S14	Combined levels	0.07454	Best estimate-only method
S15	Fire stations drive the facility half	0.07716	Partly kept
S17	Transport rebuilt from TIGER 2025	0.04906	Kept
S18	Building rebuilt from Microsoft US v2	0.05462	Dropped
S19	Fire and school gaps from USGS structures, $E = 0$	0.02996	Superseded
S20	S19 with ZBP establishments apportioned by places	0.03153	Dropped
S21	S19 with EMS from USGS ambulance services	0.04289	Dropped
S22	S19 with HIFLD schools	0.04136	Dropped
S23	A competitor's published file, uploaded as a benchmark	0.00227	Not ours
S24	S19 with ZBP establishments apportioned by HUD business ratios	0.02380	Final B
S25	S24 with the building gap against Microsoft Global ML	0.02152	Final A

S25 is the best public score of this pipeline and depends on no tuned scalar: every component is computed from a public reference. S24 is the same method with the building gap held at 0 where Overture has footprints. It differs from S25 in 5,644 tracts by small amounts and hedges the private split against the case where the Global ML reference fits the public tracts better than the private ones. S23 was a competitor's published file, uploaded once to see its score. It was not produced by this pipeline, its inputs include the organisers' removed reference files, and it is not selected.

7 Bias Scorecard reflections

The scorecard reports disparity ratios of the predicted score between strata. For the all-zero file every ratio was unavailable. Under the estimation phase (S13) the tribal ratio was 1.30 and the rural ratio 1.25. On the scorecard for S25 they are 2.85 for tribal against non-tribal tracts, 2.41 for rural against urban, 1.21 for high social vulnerability, 1.46 for high climate vulnerability, 1.47 for winter drought and 1.81 for wildfire hazard, with summer heat the one stratum where the gap is smaller (0.56). The rise comes mostly from transport in Eastern Oklahoma, where 805 of 1,192 scored tracts overlap a tribal area and the mean transport gap where defined is 0.355, against 0.123 in South-Central Texas, and from fire stations, whose gap where defined averages 0.44 in Eastern Oklahoma against 0.24 in Texas.

8 Best Bias Discovery: unnamed local roads on tribal reservations

8.1 The pattern and why the scorecard cannot see it

Every coverage measure in this challenge, and every metric on the Bias Scorecard, counts length or features. A road segment that Overture holds without a name adds as much length as a named one,

so a tract whose roads are all drawn but half unnamed scores as fully covered. The name is what emergency services use. A caller gives an address, computer-aided dispatch validates it against road centrelines that carry street names and address ranges [4], and routing tools give instructions by street name. This section shows that on federally recognised reservations a much larger share of the local street network has no name in Overture than on comparable non-tribal tracts, while the highways through the same reservations are named as often as anywhere else.

8.2 Measure and comparison

For each scored tract the measure is the share of drivable road length (Overture `roads` layer, classes motorway to unclassified, clipped to the tract) whose `names.primary` is empty. It is computed separately for the local classes (tertiary, residential, unclassified) and the named classes (motorway, trunk, primary, secondary). Tracts are tribal when the challenge's tribal overlap table links them to an American Indian, Alaska Native or Native Hawaiian area, split into legal areas (reservations and off-reservation trust land) and statistical areas (Oklahoma Tribal Statistical Areas, OTSAs).

Rurality changes naming a great deal: in South-Central Texas the unnamed share is 8 % in metropolitan tracts and 29 % in rural ones. Tribal tracts are therefore compared with non-tribal tracts in the same region and RUCA class (metropolitan, micropolitan, small town, rural), and the cell differences are averaged with weights equal to the number of tribal tracts in each cell. Intervals come from 1,000 bootstrap resamples of whole counties, which keeps neighbouring tracts that share mapping history together. All numbers come from stage 04 run `20261030-1019-04_analysis-8a838bb` on the challenge data only; county names come from the Census 2020 county code list [14].

Reproducing the finding. Road length comes from `<region>-overture-roads.parquet` (classes motorway, trunk, primary, secondary, tertiary, residential, unclassified), as geodesic metres of each segment clipped to the tract polygons in `<region>-census-tracts.parquet`; a segment is unnamed when `names.primary` is null. The tribal kind comes from `<region>-strata-tract-table.parquet`: a tract is a reservation tract when `tribal_legal` is true and a statistical-area tract when `tribal_any` is true and `tribal_legal` is false; area names come from `aiannh_name` in `<region>-tribal-tract-table.parquet`. The rurality class maps `ruca_primary` 1 to 3 to metropolitan, 4 to 6 to micropolitan, 7 to 9 to small town and 10 to rural. Tracts with no drivable road are left out, and tracts with no local road are left out of the local comparison. In each region and class cell holding both groups, the mean tribal share minus the mean non-tribal share is taken, and the cells are averaged with weights equal to their tribal tract counts. The interval resamples counties (the first five GEOID digits) with replacement 1,000 times and reports the 2.5th and 97.5th percentiles, dropping resamples in which no cell holds both groups.

8.3 Evidence

Figure 5 maps the measure. Table 5 and Figure 4 give the matched differences. On the 80 reservation tracts, local streets are unnamed 9.5 percentage points more often than on matched non-tribal tracts (95 % interval 3.5 to 14.7). The highways on the same tracts show no difference (−0.4, interval −7.3 to 5.6), so the gap sits on the streets where people live and the measure is not picking up a regional habit of leaving highways unnamed. The difference holds in each region with enough reservation tracts to test: Maricopa (37 tracts, +7.3), Eastern Washington (16, +4.5) and Eastern Oklahoma, where the Osage Reservation reaches +27.6 (9.8 to 29.6). Northern California's 13 reservation tracts point the same way with an interval that includes zero.

Table 5: Tribal minus non-tribal unnamed share of drivable length, matched within region and RUCA class, in percentage points, with 95 % county bootstrap intervals.

Tribal tracts	Roads	Tracts	Difference	Interval
Reservations, all regions	local	80	+9.5	3.5 to 14.7
Reservations, all regions	all drivable	80	+8.3	2.9 to 13.5
Reservations, all regions	named classes	79	−0.4	−7.3 to 5.6
Osage Reservation, Eastern Oklahoma	local	13	+27.6	9.8 to 29.6
Reservations, Maricopa	local	37	+7.3	3.0 to 10.9
Reservations, Eastern Washington	local	16	+4.5	0.0 to 14.8
Reservations, Northern California	local	13	+3.8	−2.9 to 10.1
OTSAs, Eastern Oklahoma	local	792	+3.0	−6.0 to 7.5

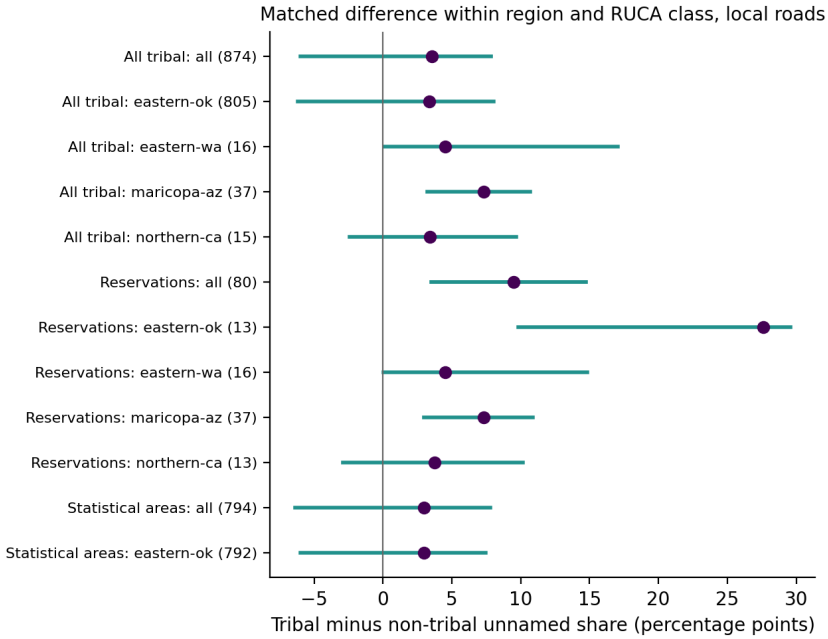


Figure 4: Matched difference in the unnamed share of local street length, tribal minus non-tribal tracts, with 95 % county bootstrap intervals. Numbers in brackets are tribal tracts.

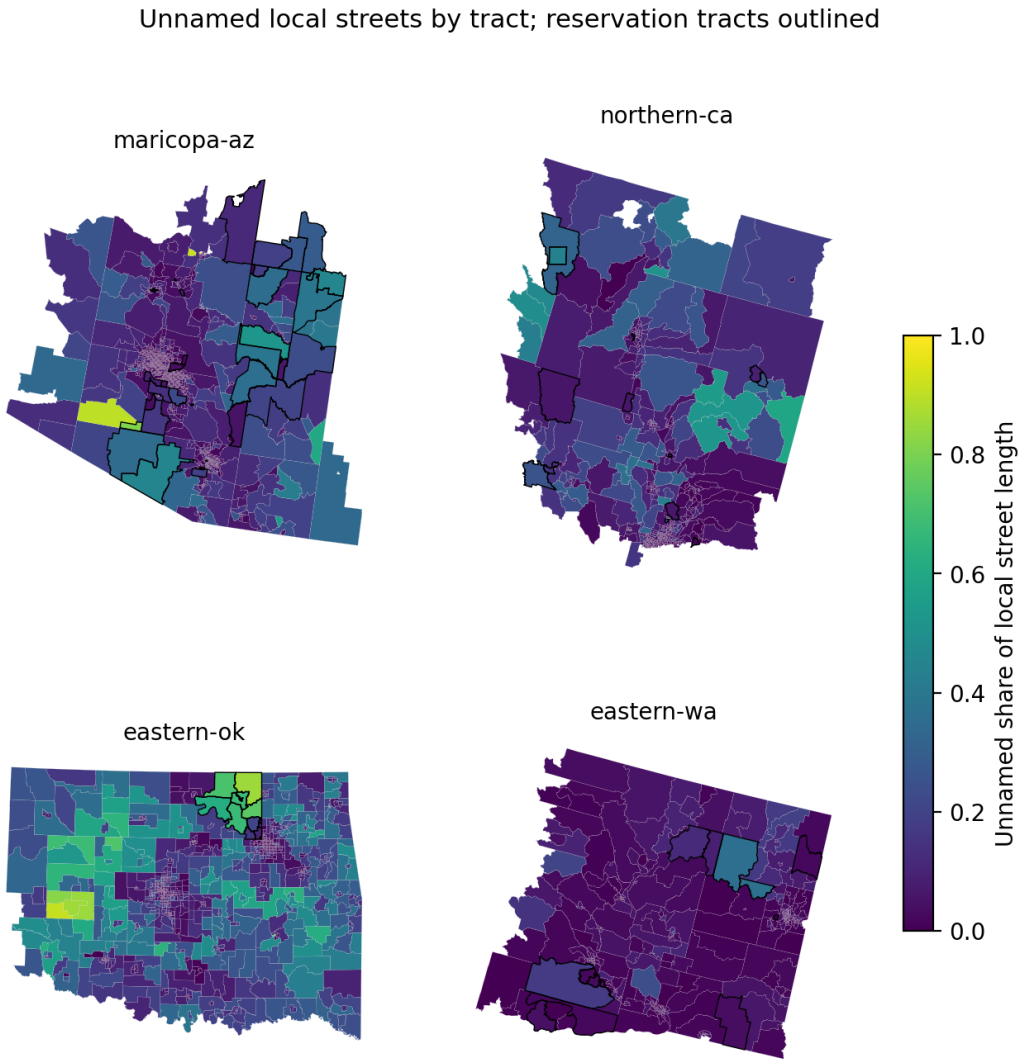


Figure 5: Unnamed share of local street length by tract (Overture 2026-08-19.0). Reservation tracts are outlined in black; the Osage Reservation is the outlined block in north-central Oklahoma. Oklahoma Tribal Statistical Areas are not outlined. Grey tracts have no local road.

The OTSAs hold the largest unnamed networks in absolute terms: 13,066 km of unnamed drivable road in the Cheyenne and Arapaho OTSA (47 % of its network) and 8,831 km in the Choctaw OTSA (34 %). After matching they do not separate from non-tribal tracts, because 792 of Eastern Oklahoma's 1,192 scored tracts lie in an OTSA and only 43 non-tribal tracts outside its metropolitan areas remain to compare with. The OTSA result is therefore unresolved by this design, and the claim made here is limited to reservations.

The 80 reservation tracts hold 294,612 residents and 8,266 km of unnamed drivable road. In 19 of them, home to 55,092 people, at least 30 % of local street length has no name. That is 24 % of reservation tracts, against 13 % of non-tribal tracts once those are weighted to the same mix of rurality.

Table 6 names the reservation tracts with the most unnamed road, with their wildfire history and social vulnerability from the challenge's strata tables. Several of them combine a largely unnamed street network with recent fire and high vulnerability. Tract 9400.04 in Osage County, Oklahoma has 78 % of its drivable length unnamed and 29 % of its land burned between 2015 and 2023. Tract 9400 in Ferry County, Washington, on the Colville Reservation, has 498 km of unnamed road, 27 % of its land burned since 2015 and an SVI percentile of 0.84. Tracts 9408 and 9411 in Pima County, Arizona, on the Tohono O'odham Nation Reservation in the Sonoran Desert, have SVI percentiles of 0.96 and a third or more of local street length unnamed.

Table 6: Reservation tracts with the most unnamed drivable road. Local: unnamed share of local street length. Burned: share of land burned 2015 to 2023 (MTBS). SVI: CDC SVI 2022 overall percentile.

GEOID	Tract	County	Reservation	Unnamed km	All	Local	Burned	SVI
40113940004	9400.04	Osage, OK	Osage	606	0.78	0.86	0.29	0.48
40113940003	9400.03	Osage, OK	Osage	524	0.62	0.62	0.12	0.74
53019940000	9400	Ferry, WA	Colville	498	0.34	0.36	0.27	0.84
04001970202	9702.02	Apache, AZ	Zuni	462	0.35	0.39	0.00	0.54
40113940011	9400.11	Osage, OK	Osage	454	0.74	0.71	0.12	0.58
06023010102	101.02	Humboldt, CA	Yurok	387	0.32	0.32	0.20	0.61
04001970502	9705.02	Apache, AZ	Fort Apache	385	0.23	0.23	0.02	0.66
04019940800	9408	Pima, AZ	Tohono O'odham	384	0.40	0.45	0.03	0.96
40113940001	9400.01	Osage, OK	Osage	327	0.61	0.68	0.37	0.83
04017940100	9401	Navajo, AZ	Fort Apache	324	0.47	0.52	0.07	0.87

8.4 Consequences

Emergency dispatch. An address on a road with no name in the map cannot be matched to a centreline, so a 911 call from it geocodes to a nearby named road, a ZIP centroid or nothing. On a reservation, response is often shared between tribal, county, state and federal agencies, and units arriving from outside lack the local knowledge that substitutes for a street name. On the Osage Reservation, where 57 % of the drivable length has no name and single tracts reach 78 %, a crew from a neighbouring county routed by a tool built on open map data receives instructions in which most turns are onto an unnamed road.

Evacuation routing. Wildfire evacuation orders name the roads to leave by and the zones they bound. Where a third of the local network is unnamed, as on the Colville and Yurok tracts with

a fifth to a quarter of their land burned since 2015, zone maps and routing products drawn from open data cannot describe the egress routes that residents of those roads would need.

Disaster relief. Damage assessment, supply distribution and welfare checks during extreme heat, which the Tohono O'odham tracts face each summer, are organised by address. Households on unnamed roads drop out of address-based lists and have to be found by coordinates.

Why the scorecard misses it. The Road Network Equity Ratio and the other scorecard metrics compare lengths and counts, and the transport component itself compares only motorway to secondary length. A reservation whose streets are drawn but unnamed has full road coverage by every one of these measures. The disparity becomes visible only through an attribute, the name, that none of them reads.

9 Limitations

The gap form in Equation 1 is taken from the organisers' description and README; their code was not available. The establishments apportionment is undocumented; the HUD business-address ratios fit it well ($t^* = 1.17$), but the organisers' crosswalk quarter and Business Patterns year are unknown. The Global ML release fits the building reference ($t^* = 1.22$), but the organisers may have used an earlier one. The TIGER 2025 and USGS 2026 files may differ from the vintages the organisers used, and their EMS reference could not be rebuilt from public data. The remaining public error of 0.0215 sits in these components. The bias discovery uses Overture names as published. A road may carry a name in a local addressing system that Overture does not hold; in that case the open map lacks the name, which is the gap being measured.

Reproducibility statement

Code, configuration and logs are in a private git repository that will be released to Zindi reviewers on request. Every compute stage runs on Kaggle from a code dataset built from a clean commit, with the commit hash written to each output. The final submissions come from commits `45a6fcc` (S25) and `30d804e` (S24) with the overrides in Table 7. Python 3.13.15 on the Kaggle image, DuckDB 1.5.4, numpy 2.1.3, pandas 2.3.3; the full `pip freeze` is saved with each run. The seed is 20261025 for every random generator. The rebuilt components involve no random draws. Both final configurations were rerun on Kaggle from later code with `make reproduce` on 30 October: S24 (run `20261030-1008-03_model-c516b52`) and S25 (run `20261030-1512-03_model-8f446f4`). Each output matched the uploaded file in every cell, with a largest float difference of 0 and no differing flag. Runtimes and memory are in Table 2. The commands are `make kaggle-run STAGE=<stage> SET="<overrides>"` for each stage in the order 01, 02, 05, 06, 07, 10, 03.

Table 7: Configuration overrides for the final submissions.

Submission	Overrides on `configs/model.yaml`
S25	`transport.source=tiger building.source=microsoft building.count=globalml poi.source=reference poi.reference_types=[fire,schools] poi.establishment_weight=bus`
S24	as S25 with `building.source=proxy building.baseline_gap=0`

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